Course Orientation & Data Science Process

What data science is and how a project runs

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Learning objectives and route through the class

By the end of this class, students should be able to
- explain what data science is, using a defensible definition, and say which disciplines it draws on;
- describe the kinds of data a data scientist meets, and state precisely what the rows and columns of a tabular data set represent;
- distinguish exploratory, predictive, and descriptive analytics, and recognize each in a worked example on the same data set;
- name and order the six stages of CRISP-DM, and state the question each stage answers;
- explain why the process is a loop rather than a pipeline; and
- locate the course's assessment, deadlines, communication channels, and required tools.

The class moves in four steps. It first motivates the field from the labour market and from the growth in data and computing. It then introduces data science through one clinical data set, using that data set to illustrate the three task families. It then presents CRISP-DM as the standard project life cycle. It closes with the course's own structure, which follows the same life cycle.

1 Why learn data science

1.1 The labour market

A ranking of the best jobs in Canada illustrates the landscape in Table 1. The point is not the precise salaries. It is that demand is spread across a family of data-related roles, and that a data role can rank highly on growth even when it does not rank highly on salary.

Table 1. Historical 2022 ranking of data-related jobs in Canada by posting growth

Rank	Job	Average salary	Growth (%) in postings, 2019–22
1	Data engineer	$85,706	242
2	Site reliability engineer	$90,010	218
3	Engineering manager	$111,515	161
...			
19	Data scientist	$76,938	60

Source: Canadian HR Reporter. Data engineering, first on this list, is the work of making data available and reliable. Much of what this course calls data preparation is the analyst's share of that work.

1.2 What is driving the field

Increasing volumes of data. Cheap and varied sensors, wider networking, and the Internet of Things all generate records that are stored by default rather than discarded.

Increasing computing power. Methods that were once impractical on a single machine now run on a laptop, which changes what is worth attempting.

Data science is also an *interdisciplinary* field with application across many domains: healthcare; finance and banking; transportation and smart cities; business and e-commerce; social media and social network analysis; and sustainability and climate. Each of these supplies both the questions and the domain knowledge without which an analysis cannot be interpreted.

Discussion prompt from the class

Why did you enrol in this course, and what do you hope to learn? Later in the class: now that you know a bit more about what data science can do, what application areas seem most interesting to you?

2 Data and tabular data

2.1 Types of data

Volumes are increasing across several different types of data.

- Tabular data.
- Databases, that is, multiple related data tables.
- Text.
- Multimedia: images, video, sound, and so on.
- Data streams and time series.

This course works with **tabular data**. That restriction is deliberate. Almost every idea in the course, from distance to model assessment, is defined first on a table, and the definitions transfer once the other data types have been given a tabular representation.

2.2 The structure of a table

Key terminology follows.

- A data set is a collection of observations.
- A data matrix of dimensions $n \times p$ contains n observations or cases, each described by p variables.
- Rows represent the cases. They are also called objects, examples, instances, or records.
- Columns represent variables. They are also called features, fields, or attributes.

A terminology warning

In an $n \times p$ data matrix there are n rows, one per object, and p columns, one per recorded variable. Throughout this course, *case* or *record* means a row and *variable* or *attribute* means a column.

2.3 The example data set

The lecture's running example is the Heart Disease data set: **12 attributes** and **918 cases**, from the UCI Machine Learning Repository, distributed through Kaggle. The first six rows are shown in Table 2.

Download and extract `heart.csv` from the Kaggle dataset page. Run the examples in order with that CSV in the working directory.

Table 2. First six rows of the Heart Disease data set

Row	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS
	numeric	category	category	numeric	numeric	category
1	40	M	ATA	140	289	0
2	49	F	NAP	160	180	0
3	37	M	ATA	130	283	0
4	48	F	ASY	138	214	0
5	54	M	NAP	150	195	0
6	39	M	NAP	120	339	0

Row	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
	category	numeric	category	numeric	category	category
1	Normal	172	N	0.0	Up	0
2	Normal	156	N	1.0	Flat	1
3	ST	98	N	0.0	Up	0
4	Normal	108	Y	1.5	Flat	1
5	Normal	122	N	0.0	Up	0
6	Normal	170	N	0.0	Up	0

One row is one patient. Seven variables are categorical and five are numeric, including the four integer-valued measurements and `Oldpeak`. These are analytical roles, not the storage types returned by an import function. `FastingBS` and `HeartDisease` encode categories as integers. `HeartDisease` is the outcome that the predictive example will try to estimate; in the full table, 508 of the 918 patients have it, which is the 55% that appears at the root of the decision tree later in this class.

R

```r
heart <- read.csv(
  "heart.csv",
  stringsAsFactors = TRUE
)

dim(heart)       # 918  12
head(heart)
str(heart)
table(heart$HeartDisease)
```

Python

```python
import pandas as pd

heart = pd.read_csv("heart.csv")

heart.shape      # (918, 12)
heart.head()
heart.dtypes
heart["HeartDisease"].value_counts()
```

Both calls return 918 rows and 12 columns. R imports text columns as factors with the option shown. Pandas text dtypes depend on the version, with a dedicated string dtype used by default in pandas 3. Neither import converts the integer-coded categories automatically. The classification example explicitly converts the R target to a factor; in Python, choosing a classifier establishes the target's role.

3 What data science is

The lecture adopts a definition from the data mining literature.

Definition

"Data mining is the analysis of (often large) observational data sets to find unsuspected relationships and to summarize the data in novel ways that are both understandable and useful to the data owner."

Hand, Mannila, and Smyth, *Principles of Data Mining*, MIT Press, 2001, page 1.

The sentence defines data mining. Data science took the definition over and widened it, and it is quoted here as written rather than with the subject swapped, because a definition a student may cite has to match its source.

Three phrases in that definition do real work. **Observational** means the researcher does not assign an experimental intervention. Observational data may be collected specifically for a research question; causal claims need extra care. Data science also uses experimental data. **Unsuspected relationships** means the analysis is expected to surprise somebody, which is only possible if the analyst is willing to be surprised. **Useful to the data owner** means the result is judged in the owner's terms, not only by a statistical criterion.

The lecture pairs the definition with a three-circle diagram. Data science sits at the intersection of computer science and information technology, mathematics and statistics, and domain or business knowledge. The pairwise intersections are themselves recognizable fields. Machine learning sits between computing and statistics, software development between computing and the domain, and traditional research between statistics and the domain.

4 Data science tasks

The lecture divides the work into three task families, summarized in Table 3.

Table 3. Three data science task families and the questions they answer

Family	Sub-tasks	Question it answers
Exploratory data analysis	summarization, visualization	What is in this table?
Predictive analytics	classification, regression	What value should I expect for an unknown case?
Descriptive analytics	clustering, anomaly detection, association rules	What structure is present, without a target?

The lecture then applies all three to the Heart Disease data.

4.1 Exploratory data analysis: an example

Figure 1 illustrates the process. The left panel is a histogram of `Age`, stacked by `HeartDisease`, with the median age marked. The second is a scatter plot of `MaxHR` against `Age`, coloured by `HeartDisease`. Together they say that patients with heart disease in this sample tend to be older and to reach a lower maximum heart rate. Nothing has been fitted; the figures describe the sample.

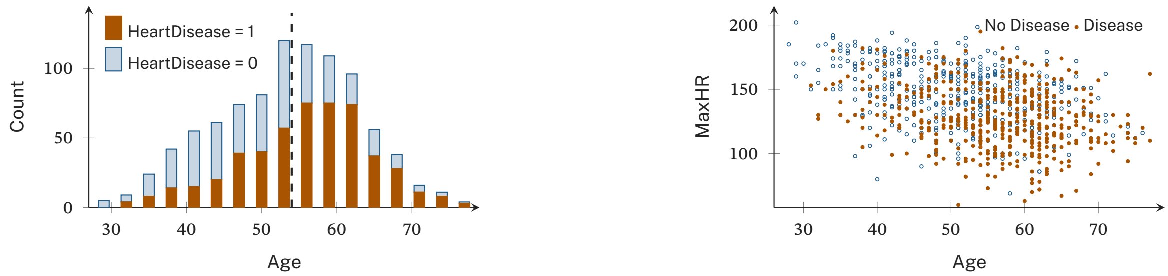

Figure 1. Exploratory views of all 918 patients. Three-year age bins start at 27.5; the dashed line marks the median age of 54. The scatter uses unjittered observations, so coincident points overlap. Neither plot fits a model.

R

```r
library(ggplot2)

ggplot(heart, aes(Age, fill = factor(HeartDisease
  ↪ ))) +
  geom_histogram(binwidth = 3, boundary = 27.5,
                 closed = "left") +
  geom_vline(
    xintercept = median(heart$Age),
    linetype = "dashed"
  )

ggplot(heart, aes(Age, MaxHR,
                  colour = factor(HeartDisease)))
  ↪ +
  geom_point(alpha = 0.6)
```

Python

```python
import matplotlib.pyplot as plt
import seaborn as sns

sns.histplot(
    heart, x="Age", hue="HeartDisease",
    bins=[27.5 + 3*i for i in range(18)],
    multiple="stack"
)
plt.axvline(heart["Age"].median(), ls="--")
plt.show()
plt.figure()
sns.scatterplot(
    heart, x="Age", y="MaxHR",
    hue="HeartDisease", alpha=0.6
)
```

The calls use the same three-year bin edges and all 918 patients. The library defaults differ: `ggplot2` stacks histogram bars by default, whereas `seaborn` layers them unless you ask for `multiple="stack"`.

4.2 Predictive analytics: an example

A classification tree with `HeartDisease` as the target appears in Figure 2. Each node in the printed tree carries three values: the predicted class, the proportion of disease cases in the node, and the share of the training data that reaches the node.

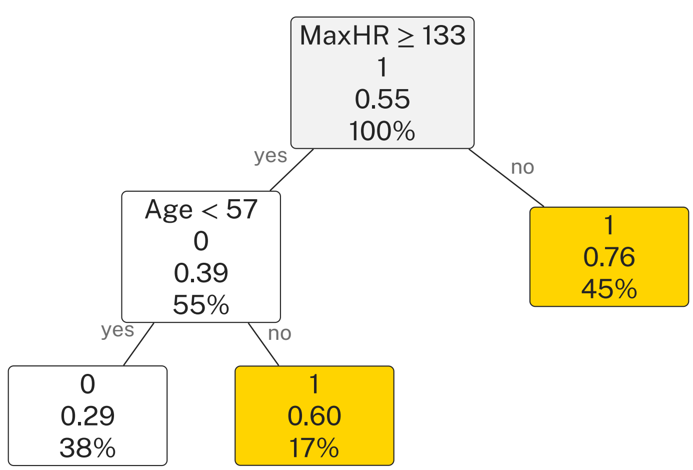

Figure 2. A depth-two classification tree for `HeartDisease`. Each node gives the predicted class, the proportion of disease cases, and the share of the training data reaching it. Shaded leaves predict disease.

Read the tree as a set of rules. A patient whose maximum heart rate is below 133 is predicted to have heart disease, and 76% of the patients in that leaf do. Above that threshold, age decides. Below 57 the prediction is no disease, at 57 or above it is disease. The two splits cut the `Age–MaxHR` plane into three rectangles, shown in Figure 3. The fitted boundaries are 132.5 and 56.5, equivalent to these integer rules on the recorded measurements. The training accuracy is **71.24%**, or 654 correct predictions out of 918.

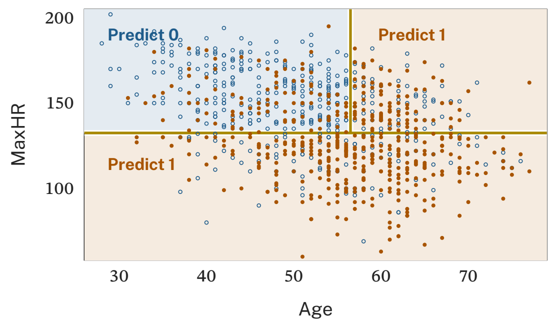

Figure 3. The same fitted tree partitions the `Age–MaxHR` plane. Blue regions predict no disease and rust regions predict disease; the points show the observed classes. The gold boundaries represent the two fitted splits.

R

```r
library(rpart)

fit <- rpart(
  factor(HeartDisease) ~ Age + MaxHR,
  data = heart, method = "class",
  control = rpart.control(
    maxdepth = 2, cp = 0.01, xval = 0)
)
plot(fit)
text(fit, use.n = TRUE)

mean(predict(fit, type = "class") ==
     factor(heart$HeartDisease))
```

Python

```python
from sklearn.tree import (
    DecisionTreeClassifier, plot_tree
)

X = heart[["Age", "MaxHR"]]
y = heart["HeartDisease"]

fit = DecisionTreeClassifier(
    max_leaf_nodes=3, random_state=3141
).fit(X, y)
plt.figure(figsize=(9, 4))
plot_tree(fit, feature_names=X.columns,
          class_names=["0", "1"], filled=True)
plt.show()
fit.score(X, y)
```

The controls are explicit. R limits depth and applies its complexity parameter; Python limits the tree to three leaves. These settings yield the same rules and predictions for this dataset. They are not generally interchangeable settings, and other datasets may produce different trees. Both methods use Gini impurity.

Common misconception

71.24% is a *training* accuracy, because the tree is scored on the same rows that chose its splits. Training performance is often optimistic; it is neither a guarantee nor a mathematical upper bound on future performance. The course returns to this when it covers performance estimation methods; until then, treat any accuracy measured on training data as a diagnostic, not a result.

4.3 Descriptive analytics: an example

The code below runs k-means on the same two variables and flags the five observations furthest from their assigned cluster centre. No target is used. For the fixed initialization below, the ranked CSV row numbers are 830, 3, 508, 391, and 569, excluding the header. Their (Age, MaxHR) pairs are (29, 202), (37, 98), (40, 80), (51, 60), and (38, 105). Figure 4 locates these cases among all observations.

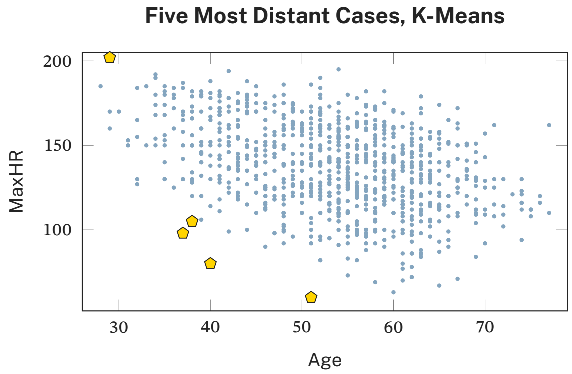

Figure 4. The five cases furthest from their assigned k-means centre are marked with gold pentagons. Distances are calculated after population-SD scaling; the axes retain the original units. A flagged case needs investigation, and is not automatically an error.

R

```r
X <- as.matrix(heart[c("Age", "MaxHR")])
Z <- scale(X, scale = FALSE)
Z <- sweep(Z, 2, sqrt(colMeans(Z^2)), "/")
km <- kmeans(
  Z, centers = Z[c(1, 301, 601), ],
  algorithm = "Lloyd", iter.max = 300
)

d <- sqrt(rowSums(
  (Z - km$centers[km$cluster, ])^2
))
order(-d, seq_along(d))[1:5]
```

Python

```python
import numpy as np
from sklearn.cluster import KMeans
X = heart[["Age", "MaxHR"]].to_numpy()
Z = (X - X.mean(0)) / X.std(0, ddof=0)
km = KMeans(
    n_clusters=3, init=Z[[0, 300, 600]],
    n_init=1, algorithm="lloyd",
    max_iter=300, tol=0
).fit(Z)

d = np.linalg.norm(
    Z - km.cluster_centers_[km.labels_], axis=1)
np.lexsort((np.arange(len(d)), -d))[:5] + 1
```

Both versions center and divide by population standard deviations, then run Lloyd updates from the same three patient rows. This avoids differences between R's default sample-SD scaling and Python's population-SD scaling, and between their random initialization routines. Ties are resolved by CSV row order. The choice of variables, scaling, and initialization affects the ranking. The fixed initialization is for a reproducible teaching example; other starting centres can lead to other local solutions.

Interpretation guidance

A flagged case is not a wrong case. Anomaly detection produces a ranked list for human attention. Whether an unusual maximum heart rate is a data-entry error, a rare patient, or the most clinically interesting record in the table is a question the method cannot answer.

Check your understanding

Which of the three examples used a labelled target, and which did not? Which one could still be run if the `HeartDisease` column were deleted?

5 CRISP-DM: the typical data science process

CRISP-DM is the **CR**oss **I**ndustry **S**tandard **P**rocess for **D**ata **M**ining. In the Data Science Process Alliance's 2020 convenience poll of 109 respondents, CRISP-DM received 49%, followed by Scrum, Kanban, and "my own" at 18%, 12%, and 12%. The remaining responses were Microsoft TDSP (4%), Other (3%), None (2%), and SEMMA (1%). The eight reported percentages sum to 101% after rounding. This self-selected poll does not estimate prevalence among all practitioners.

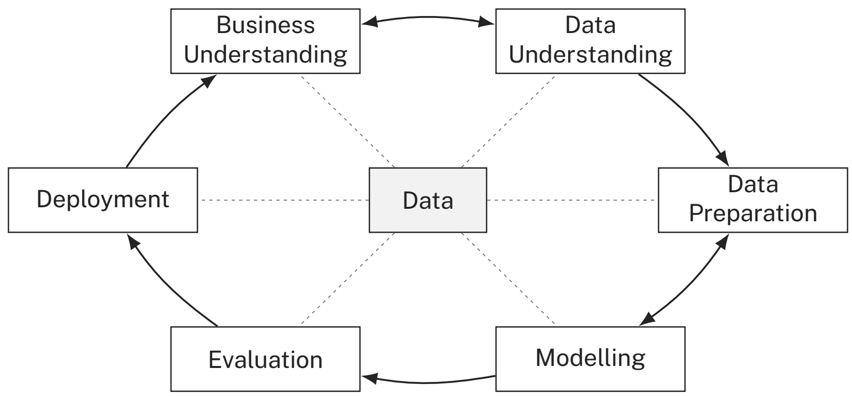

Figure 5. The six CRISP-DM stages. The cycle runs clockwise from business understanding, the two double-headed arrows mark the pairs that are worked back and forth, and every stage reads from and writes to the same data.

The six stages and their associated questions, shown in Figure 5, are as follows.

1. **Business understanding.** What are the business goals? How can data science help achieve them?
2. **Data understanding.** What data do we have or need? Are there data quality issues?
3. **Data preparation.** How do we clean and transform data for modelling?
4. **Modelling.** What modelling techniques should we use? How do we calibrate them?
5. **Evaluation.** Does the selected solution meet the business goals?
6. **Deployment.** How do we deliver our solution to the business?

Two features of the diagram carry most of the meaning. The double-headed arrows between business and data understanding, and between data preparation and modelling, record that these pairs are negotiated together, because the choice of model determines what the prepared table must look like, and what the data can support constrains what goal is realistic. The outer ring records that the process repeats; evaluation returns to business understanding rather than terminating the project.

Discussion prompts from the class

At the data understanding stage: can you think of any potential data quality issues or limitations? At the deployment stage: what kind of deliverables would provide value to stakeholders?

Think, pair, share

Take a minute to think about a concrete data science application you are curious about. What data would you need, and do you think it would be readily available? What challenges do you expect to face? Discuss with your neighbour, then volunteer to share with the class.

6 About the course

6.1 The syllabus map

The tentative syllabus is drawn as one map whose spine is CRISP-DM. Business understanding, data understanding, data preparation, data modelling, evaluation, and deployment run left to right, with evaluation feeding back to business understanding. Three of those stages carry the course's technical content, listed in Table 4.

Table 4. Course topics mapped to CRISP-DM stages

Stage	Topics hanging from it
Data understanding	importing data; summarizing; visualizing
Data preparation	cleaning; transforming; feature engineering; reducing dimensionality
Data modelling, predictive	tasks: classification, regression. Models: linear models, support vector machines, tree-based models. Assessment: performance metrics, performance estimation methods
Data modelling, descriptive	distance functions; clustering, split into partition-based and hierarchical clustering; outlier detection
Deployment	reproducible reports; optional Shiny applications in R or Python

6.2 Learning outcomes for the course

By the end of the course, you should be able to

1. describe the main steps in a data science project;
2. transform data to prepare it for analysis and modelling;
3. explore data using appropriate statistics and visualizations;
4. design and implement a data science application, using predictive and descriptive modelling techniques;
5. evaluate, compare, and select models for a dataset; and
6. prepare reproducible reports in R Markdown or Python notebooks.

6.3 Technologies

Prepare one working R or Python environment. A free cloud notebook is sufficient, using Posit Cloud or Kaggle for R, or Posit Cloud, Kaggle, or Google Colab for Python. For local work, install R from CRAN with an editor such as RStudio, or Python 3 with a virtual environment and an editor such as VS Code or JupyterLab. Both languages use the same assessment requirements and rubric. Contact the instructor early if computer access is a barrier. The syllabus's reporting outcome is R Markdown or Python notebooks. Shiny applications in either language are an optional extension, not an additional assessed requirement.

The course toolchains include `tidyverse`, `rpart`, and `cluster` in R, and `pandas`, `seaborn`, and `scikit-learn` in Python. Free cloud services have usage caps. Save your files regularly and record the dependencies needed to reproduce the analysis. Keep assessed notebooks private until the deadline and the instructor opens sharing.

6.4 Communication

- Announcements are posted in the course GitHub repository, `https://github.com/MAPS-Lab/CSCI-3141`. Course materials are public and can be read without signing in. You need a GitHub account for your assessment portfolio. Keep assessed work private until its deadline and the instructor opens sharing. Check the repository and your institutional email regularly.
- Use Microsoft Teams for class questions and office-hour appointments.
- For private matters, use e-mail, with a subject line of the form `[CSCI 3141] Subject matter`.
- Instructor is Prof. Gabriel Spadon, `spadon@dal.ca`. Replies are normally within three business days.

Office hours are held **by appointment through Microsoft Teams**. Write ahead with the subject you would like to discuss. Individual and small-group appointments are available. Bring a focused question and ask early rather than waiting until a deadline.

6.5 Classes

Classes run Tuesdays and Thursdays, **16:05–17:25**, in Carleton Tupper Building room **3H01**, in person, across the twelve teaching weeks of the term. The syllabus and the course calendar carry this term's dates. Attendance is not mandatory, but it is recommended, because classes include active-learning activities such as in-class response questions and work in pairs or small groups, which do not reproduce well from slides alone. Course materials are published in the course GitHub repository. There is no separate lab section; the integrated lab work begins in the Thursday classes.

6.6 Academic integrity and conduct

Academic integrity is expected from all students at Dalhousie. Review the Academic Integrity resources linked from the course GitHub repository. Originality checking follows syllabus A.9.1. To use its equivalent originality-checking alternative without penalty, notify the instructor by September 22. Everyone at Dalhousie should be treated with dignity and respect, in class and online.

Generative AI

Generative AI is **prohibited on integrated labs** and examinations, and on assignments unless its use is expressly authorized in the assignment instructions. When it is authorized, follow the stated disclosure requirements.

6.7 Assessment

Table 5 lists the course assessment components, their weights, and the rule applied to each.

Passing requires three separate thresholds. The weighted total, the combined assignment component, and the final examination must each reach 50%.

Integrated labs are marked with the common course rubric, and the best 10 of 12 results contribute 15% of the course grade. Submit each lab by 23:59 two calendar days after it begins. No quizzes are scheduled or graded. Assignments are individual work, due at 23:59 on the announced date. Assignments 1 to 3 lose **5 percentage points per late calendar day**, up to seven days, after which later work requires an approved absence or accommodation process. Assignment 4 has no routine late window. Use the syllabus's equivalent private repository when accommodation, safety, privacy, copyright, or licensing requires it.

Table 5. Course assessment components and their weights

Component	Weight	Rule
Integrated labs	15%	best 10 of 12 results
Assignment 1	7.5%	import data and perform basic data exploration; due 11 October 2026
Assignment 2	7.5%	pre-process and explore data; due 1 November 2026
Assignment 3	15%	design and implement a predictive modelling solution; due 29 November 2026
Assignment 4	15%	design and implement a descriptive modelling solution; due 9 December 2026
Midterm	15%	in class, 20 October 2026
Final examination	25%	cumulative, scheduled by the Registrar
Total	**100%**	

The midterm takes place in class on October 20, from 16:05 to 17:25, and the Registrar schedules the cumulative final examination. Bring valid Dalhousie photo ID. Course and personal notes are allowed unless the examination lists narrower approved aids. Internet access, communication, and generative AI are prohibited.

Build the portfolio in a private work repository with instructor access. Commit regularly, then submit the full commit SHA and its commit URL through the announced private channel by the deadline. A branch link or movable tag alone does not fix the submitted version. Keep cloud notebooks and answer-revealing links private too. After the deadline and when the instructor opens sharing, publish that same commit and its history in your public portfolio and share that version in class.

Preserve assessment history permanently. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded commits, and do not move submitted tags. Keep submitted commits reachable. Record corrections as new commits, retaining the original submission. Git author dates alone are not deadline evidence; the submission receipt and instructor-retained snapshot provide the independent reference. Contact the instructor if an accidental disclosure requires correction.

Course materials released by the instructor may be shared with credit to the author and CSCI 3141 unless explicitly marked private or restricted. Preserve source links, third-party attributions, and licence terms. This permission does not authorize sharing another student's work or releasing assessed work early.

For an eligible short-term absence, contact the instructor before the requirement and submit the declaration within three calendar days after the last day of the absence, at most twice per course. Follow syllabus A.7.3 for other absences.

6.8 Dates

Table 6 lists the key course dates and deadlines for fall 2026.

Table 6. Key course dates and deadlines for fall 2026

Date, 2026	Event	Kind
8 September	Classes begin	course
22 September	Last day to add	University
7 October	Last day to drop without a W	University
11 October	Assignment 1 due	course
20 October	Midterm, in class	course
1 November	Assignment 2 due	course
5 November	Last day to drop with a W	University
9–13 November	Fall Break	University
29 November	Assignment 3 due	course
3 December	Last class	course
9 December	Assignment 4 due	course
Registrar	Final examination	University

6.9 Course text and resources

There is **no required paid textbook**. The lecture slides, notes, and integrated lab handouts in the course GitHub repository are the primary material. Cheatsheets and documentation links are posted with each module. The following texts are optional further reading, with the same editions and guidance as the slides. Check Dalhousie Libraries for access before obtaining a copy.

Rigdon, S. E., Fricker, R. D., and Montgomery, D. C., *Introduction to Probability and Statistics for Data Science with R*, Cambridge University Press, 2024.

Shah, C., *A Hands-On Introduction to Data Science with Python*, 2nd edition, Cambridge University Press, 2026.

Torgo, L., *Data Mining with R: Learning with Case Studies*, 2nd edition, CRC Press, 2017. Earlier editions are acceptable.

6.10 This week

Read the syllabus, check course repository and Teams access, and prepare one working local or cloud environment in R or Python before Thursday. Submit the ungraded syllabus and integrity acknowledgement by September 22. Thursday starts Lab 01. The class begins with a summary-statistics warm-up, then you choose one listed Atlantic environmental dataset and establish its descriptive baseline.

7 Summary

Data science combines computing, statistics, and domain knowledge to learn from observational and experimental data. This course uses tabular data, with n cases as rows and p attributes as columns. The Heart Disease example has 918 cases and 12 attributes.

Exploratory analysis summarizes and visualizes the sample. Predictive analytics learns a target, as in the tree splitting `MaxHR` at 133 and `Age` at 57. Its 71.24% training accuracy does not establish future performance. Descriptive analytics finds structure without a target, as in the k-means example that flags five patients for investigation.

CRISP-DM connects business understanding, data understanding, data preparation, modelling, evaluation, and deployment. Later evidence can revise earlier decisions. The course follows this process, and its four assignments move from import to prepare to predict to describe. Consult the syllabus for assessment and integrity rules, and keep portfolio work private until its deadline and the instructor's release.